Flood Risk Assessment using ArcGIS Pro

1. Introduction

This project demonstrates a GIS-based flood risk assessment for a study area in South Australia using ArcGIS Pro. Spatial datasets were processed and combined to identify areas with different levels of flood susceptibility.

2. Objectives

• Analyse terrain characteristics.
• Assess proximity to rivers.
• Incorporate land cover information.
• Produce a final flood risk map.

3. Data

Digital Elevation Model (DEM)
River network
Land cover dataset
Roads and Local Government Area boundaries

4. Methodology

Generate DEM-derived slope.
Calculate flow accumulation.
Create Euclidean distance to rivers.
Reclassify all input rasters.
Combine datasets using Weighted Overlay.
Produce the final flood risk map.

5. Results

The resulting flood risk map classifies the study area into relative flood risk categories. Areas close to rivers, with lower slopes and favourable drainage characteristics, generally showed higher flood susceptibility.

6. Software

ArcGIS Pro
Spatial Analyst
Raster Analysis
Weighted Overlay

7. Conclusion

This project demonstrates practical GIS and raster analysis skills for environmental hazard assessment and supports spatial decision making.